

Exp No. 9: Massive MIMO Capacity vs SNR and Gain Using MATLAB

AIM: To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

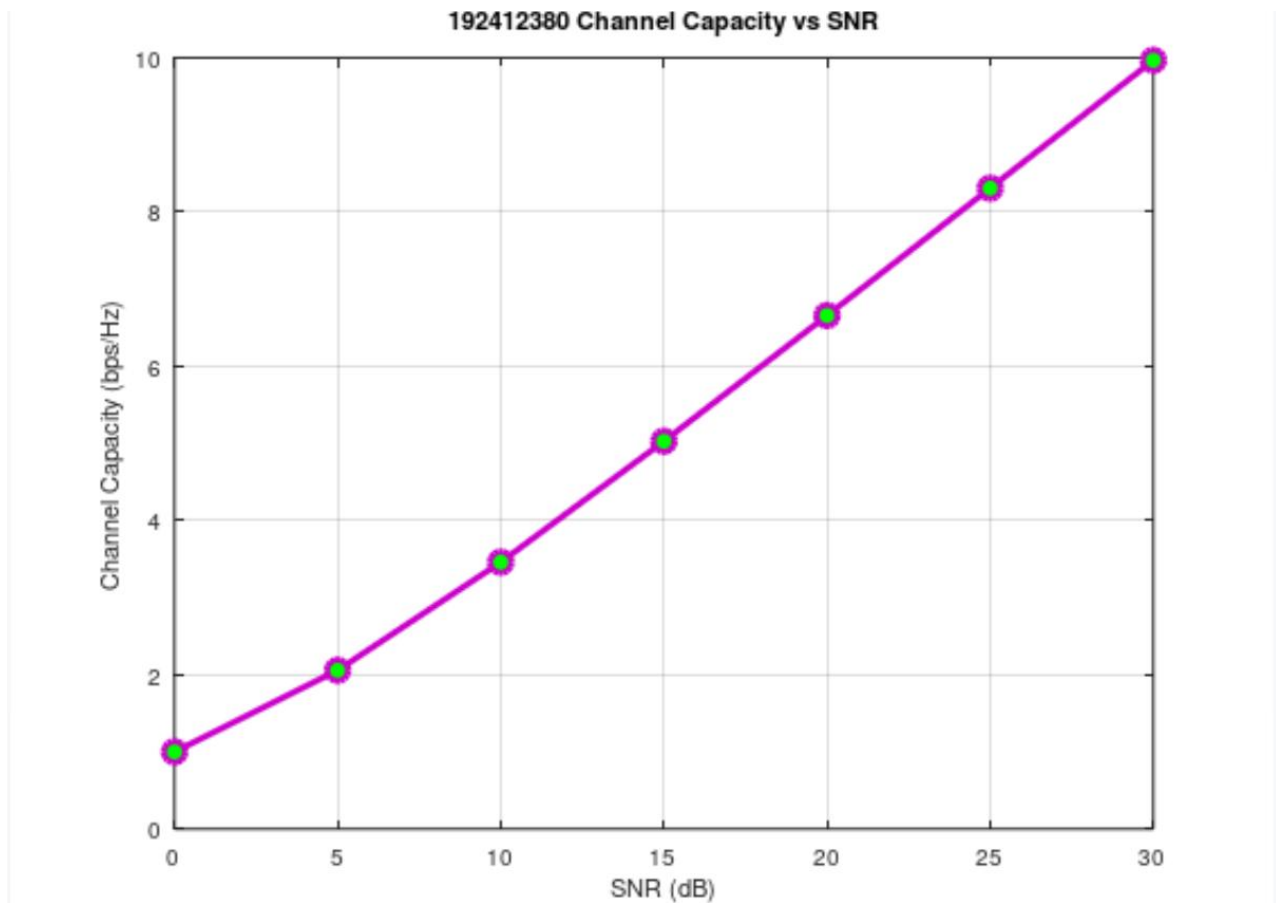
Objective 1: Analyze Channel Capacity (bps/Hz)

Analyze the channel capacity of a Massive MIMO system for different signal-to-noise ratio (SNR) values using MATLAB. Observe how channel capacity increases with improving signal quality based on the Shannon capacity principle.

Program :

```
1  clc;
2  clear;
3  close all;
4
5  % SNR Range (dB)
6  snr = 0:5:30;
7
8  % Convert SNR to Linear Scale
9  snr_linear = 10.^(snr/10);
10
11 % Channel Capacity (bps/Hz)
12 capacity = log2(1 + snr_linear);
13
14 figure;
15
16 % Plot with Different Colors
17 plot(snr, capacity, ...
18      '-o', ...
19      'Color', [0.8 0 0.8], ...      % Purple line
20      'MarkerFaceColor', [0 1 0], ... % Green markers
21      'Linewidth', 2, ...
22      'Markersize', 7);
23
24 grid on;
25
26 xlabel('SNR (dB)');
27 ylabel('Channel Capacity (bps/Hz)');
28 title('192412380 Channel Capacity vs SNR');
29
30 drawnow;
31 pause(5);
```

OUTPUT :



Objective 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the signal-to-noise ratio for different communication scenarios using MATLAB. Analyze the effect of SNR on the quality and reliability of wireless communication in Massive MIMO systems.

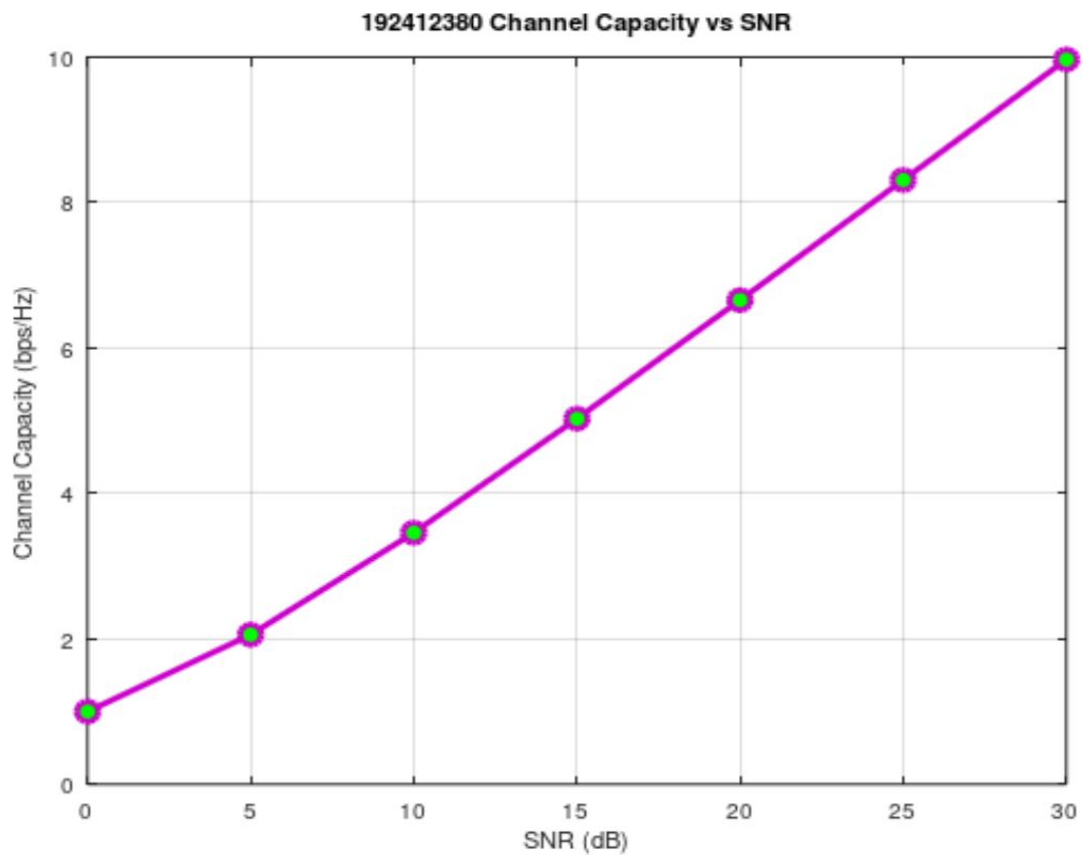
Program

```

1  clc;
2  clear;
3  close all;
4
5  % SNR Range (dB)
6  snr = 0:5:30;
7
8  % Convert SNR to Linear Scale
9  snr_linear = 10.^(snr/10);
10
11 % Channel Capacity (bps/Hz)
12 capacity = log2(1 + snr_linear);
13
14 figure;
15
16 % Plot with Different Colors
17 plot(snr, capacity, ...
18      '-o', ...
19      'Color', [0.8 0 0.8], ...      % Purple line
20      'MarkerFaceColor', [0 1 0], ... % Green markers
21      'LineWidth', 2, ...
22      'MarkerSize', 7);
23
24 grid on;
25
26 xlabel('SNR (dB)');
27 ylabel('Channel Capacity (bps/Hz)');
28 title('192412380 Channel Capacity vs SNR');
29
30 drawnow;
31 pause(5);

```

OUTPUT :



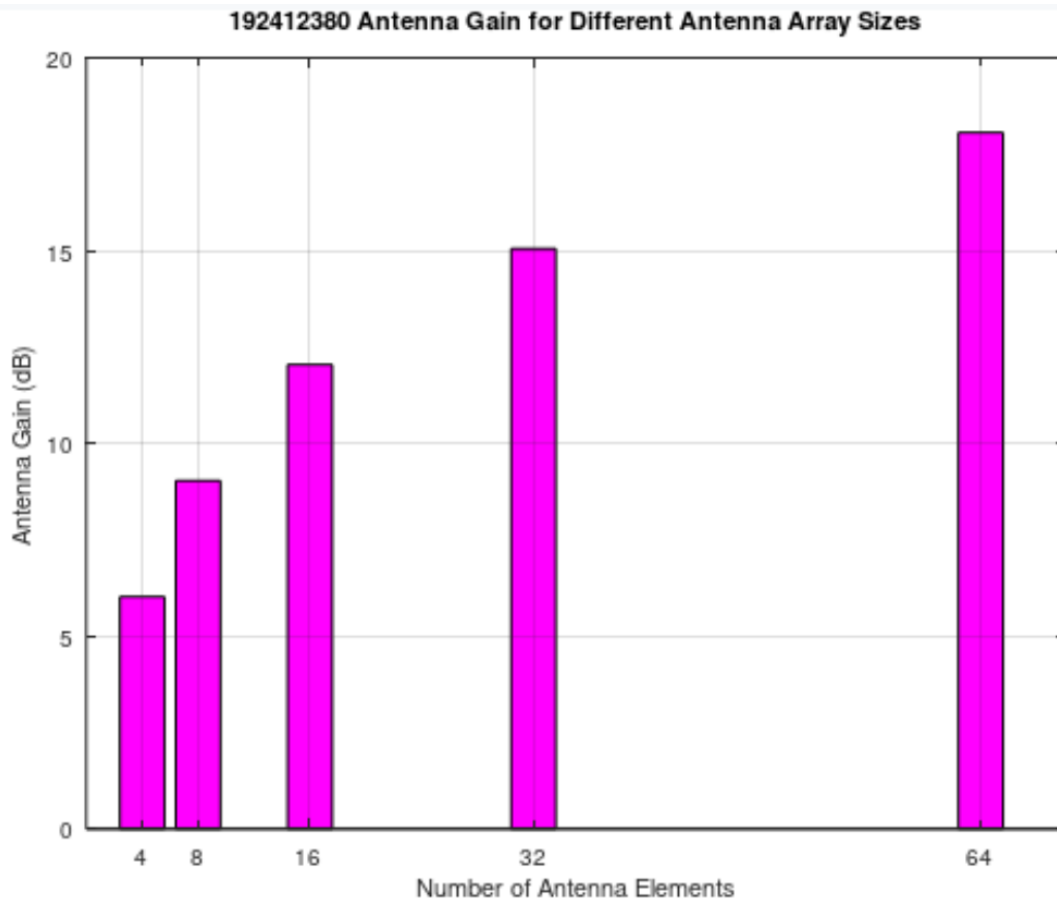
Objective 3: Analyze Antenna Gain (dB)

Analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB. Observe how increasing the antenna array size improves directional gain and enhances communication performance.

program

```
1  clc;
2  clear;
3  close all;
4
5  % Number of Antenna Elements
6  antennas = [4 8 16 32 64];
7
8  % Antenna Gain (dB)
9  gain = 10 * log10(antennas);
10
11 figure;
12
13 % Bar Graph (Magenta Color)
14 bar(antennas, gain, 'm');
15
16 grid on;
17
18 xlabel('Number of Antenna Elements');
19 ylabel('Antenna Gain (dB)');
20 title('192412380 Antenna Gain for Different Antenna Array Sizes');
21
22 drawnow;
23 pause(5);
```

OUTPUT :



Result:

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.

Practical Questions

A wireless communication system employing **Massive MIMO technology** enhances network performance by using a large number of antenna elements to improve channel capacity, increase signal-to-noise ratio (SNR), and provide higher antenna gain. These parameters play a crucial role in achieving high data rates and spectral efficiency in 5G and beyond networks. Using MATLAB, simulate a Massive MIMO system and analyze its performance.

1. To analyze the variation of channel capacity (bps/Hz) with respect to Signal-to-Noise Ratio (SNR) in a Massive MIMO system using MATLAB.
2. To compare the Signal-to-Noise Ratio (SNR) values for different communication conditions and evaluate their impact on system performance.
3. To analyze the antenna gain achieved by varying the number of antenna elements in a Massive MIMO system using MATLAB.